

ABSTRACT

"Physioplay" is a revolutionary mobile app developed using machine learning-based pose detection and the Flutter framework, aimed at providing physiotherapy for children with Autism Spectrum Disorder (ASD) and Cerebral Palsy (CP). By harnessing advanced pose detection algorithms, the app accurately tracks users' movements. Through an array of interactive games, "Physioplay" offers tailored exercises to enhance motor skills, coordination, and cognitive development, all while ensuring correct posture through real-time feedback.

Intuitively designed with a user-friendly interface, the app engages children in therapeutic activities that align with specific goals. The gamified experience not only boosts engagement but also motivates users to adhere to their therapy routines. The seamless integration of machine learning and Flutter enables cross-platform accessibility, making the app available to both Android and iOS users, while also allowing for continuous updates and enhancements.

Extensive testing involving ASD and CP children, coupled with expert input from physiotherapists, validates the app's effectiveness. Results show improved engagement, adherence, and therapeutic progress. "Physioplay" stands as a transformative solution in pediatric physiotherapy, combining machine learning and gamification through Flutter to offer an innovative approach that positively impacts the development and well-being of children facing ASD and CP challenges.

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1. INTRODUCTION

1.1 Background

Autism spectrum disorder (ASD) and cerebral palsy (CP) are complex neurodevelopmental conditions that affect a significant number of children globally. Children with ASD may experience challenges in communication, social interaction, and repetitive behaviors, while those with CP face difficulties in motor function, muscle control, and coordination. Physiotherapy plays a crucial role in improving the physical and cognitive abilities of these children, helping them achieve greater independence and a better quality of life.

Traditional physiotherapy interventions involve one-on-one sessions with therapists, but these can often be tedious and repetitive for young kids, leading to disinterest and limited progress. To address this issue, the idea of developing an innovative app, known as PhysioPlay, emerged. The goal was to leverage the power of technology, gamification, and machine learning to create an interactive and personalized physiotherapy experience for children with autism and cerebral palsy.

1.2 Problem Statement

Children diagnosed with Autism Spectrum Disorder (ASD) and Cerebral Palsy (CP) face challenges in motor skills and cognitive development. Traditional physiotherapy approaches lack engagement and accessibility, hindering therapeutic progress. The "Physioplay" project aims to address this by developing a mobile app that utilizes machine learning pose detection and Flutter to create interactive games, offering real-time feedback and a motivating framework for therapy.

The goal is to enhance engagement, adherence, and therapeutic outcomes for ASD and CP children, ultimately improving their quality of life through an innovative and accessible physiotherapy solution.

1.3 Product Scope

The "Physioplay" app will be developed as a user-friendly mobile application employing machine learning pose detection and Flutter. It aims to provide engaging and effective physiotherapy exercises for children with Autism Spectrum Disorder (ASD) and Cerebral Palsy (CP). The app will feature a diverse array of interactive games targeting specific motor skills and cognitive goals, incorporating real-time pose detection to ensure correct movements. Users and caregivers will access personalized profiles for progress tracking, benefiting from gamification elements that motivate participation and achievements. The app will be available on Android and iOS platforms, while expert collaboration will guarantee exercise alignment with therapeutic practices. Regular updates will introduce new features, ensuring the app remains dynamic and tailored to evolving user needs.

1.4 Purpose

The "Physioplay" app aims to transform pediatric physiotherapy for children with ASD and CP by leveraging machine learning pose detection and Flutter technology. Through interactive games, the app strives to enhance engagement and adherence to therapy exercises, making the process enjoyable and motivating. Tailored to individual needs, "Physioplay" offers personalized exercises targeting specific developmental goals, while real-time pose detection ensures correct movements. The app's accessibility on Android and iOS platforms facilitates widespread usage, and collaboration with experts guarantees evidence-based practices. By continuously evolving based on user feedback and advancements in therapy, "Physioplay" aspires to provide an innovative and effective solution to improve the well-being and developmental outcomes of its users.

2. OBJECTIVES

The "Physioplay" app aims to engage and motivate children with ASD and CP through interactive games, utilizing machine learning pose detection for real-time feedback to ensure correct exercise execution. The app's personalized approach tailors exercises to individual needs, enhancing adherence and making therapy enjoyable. Collaborating with experts, it ensures therapy alignment with best practices. With cross-platform accessibility, users can track progress, fostering motivation. Regular updates based on feedback and evolving therapies ensure the app's continual improvement. Ultimately, the app seeks to empower users, enhancing their motor skills, coordination, and cognitive development while improving their overall quality of life.

3. LITERATURE REVIEW

Existing literature highlights the adoption of technology, including machine learning and gamification, to enhance pediatric physiotherapy for children with ASD and CP. Research emphasizes the significance of engaging interventions for improved adherence and outcomes. Pose detection technology has shown promise in providing real-time feedback for exercise execution, aligning with the "Physioplay" app's approach.

Gamification's motivational impact on therapy participation is well-documented, supporting the app's incorporation of rewards and interactive game elements. Personalized and adaptable interventions have demonstrated positive effects on motor skills and cognitive development for children with ASD and CP, aligning with the app's emphasis on tailored exercises.

Cross-platform accessibility is recognized as essential for diverse user engagement, a principle reflected in the app's design. In essence, literature highlights the potential of combining machine learning, gamification, and personalized therapy in the "Physioplay" app to contribute to innovative interventions for enhancing the well-being of children with ASD and CP.

4. SOFTWARE REQUIREMENT ANALYSIS

4.1 Functional requirements

4.1.1. User Registration and Profiles

The app should allow users to create individual accounts with comprehensive profiles, including essential details such as the child's diagnosis, age, and therapy goals. Profiles should be customizable to accommodate multiple children if needed, ensuring personalized and relevant experiences.

4.1.2 Interactive Game Library

A diverse selection of interactive games, specifically designed to enhance motor skills, coordination, and cognitive development, should be available within the app. Games should be categorized based on their therapeutic objectives and difficulty levels, offering engaging and intuitive experiences suitable for children with ASD and CP.

4.1.3 Machine Learning Pose Detection

Integration of advanced machine learning algorithms for real-time pose detection is essential. The system should provide instant feedback on exercise execution, guiding users to maintain correct movements and postures, contributing to the effectiveness and safety of the therapy sessions.

4.1.4 Gamification Elements

The app should incorporate gamification elements, such as rewards, achievements, and virtual badges, to motivate consistent participation. User progress and achievements should be tracked and visualized, allowing users and caregivers to monitor accomplishments and maintain motivation.

4.1.5 Customization and Adaptation

A dynamic system that tailors exercises according to each user's individual needs, developmental levels, and therapy objectives is necessary. Users or caregivers should have the flexibility to adjust game settings and difficulty levels, ensuring a personalized and adaptive therapy experience.

4.1.6 Progress Tracking and Reporting

The app should maintain a comprehensive record of user progress, including completed exercises, achievements, and performance improvements over time. Regular reports or summaries should be generated, providing valuable insights for caregivers and healthcare professionals to monitor and adjust therapy plans.

4.1.7 Cross-Platform Compatibility

Development for both Android and iOS platforms is crucial to ensure broad accessibility. The app should offer a consistent and seamless user experience across different devices, enabling wider adoption and engagement.

4.1.8 Expert Collaboration and Content Review

Collaboration with physiotherapists and healthcare experts is vital. The app should facilitate expert review and contribution of exercise content, ensuring alignment with established best practices and allowing for the integration of new exercises or modifications as needed.

4.1.9 Continuous Updates and Enhancements

The app's design should support regular updates and improvements. A modular architecture should allow for the introduction of new games, features, and refinements based on user feedback, emerging research, and evolving therapeutic approaches, ensuring the app remains current and effective.

4.2 Nonfunctional requirements

4.2.1 Usability and User Experience (UX)

The app should provide an intuitive and user-friendly interface, catering to the cognitive abilities and sensory sensitivities of children with ASD and CP. Navigation, game controls, and feedback mechanisms should be designed to ensure a positive and engaging user experience.

4.2.2 Performance and Responsiveness

The app must exhibit minimal latency and responsiveness, delivering real-time feedback during exercises. It should function smoothly on various devices, maintaining consistent performance and pose detection accuracy.

4.2.3 Security and Data Privacy

Strong security measures should be implemented to safeguard user data, including personal and therapy-related information. User accounts and sensitive data should be encrypted, and compliance with data protection regulations should be ensured.

4.2.4 Scalability and Load Handling

The app's infrastructure should be designed to handle a growing user base and potential spikes in usage, ensuring consistent performance even during peak times.

4.2.5 Interoperability and Integration

The app should be designed to seamlessly integrate with external systems, such as electronic health records or therapy management platforms, enabling healthcare professionals to access and analyze user data.

4.2.6 Cross-Device Compatibility

The app should be compatible with a range of devices, screen sizes, and orientations, ensuring that users can engage with the app on their preferred device without limitations.

4.2.7 Continuous Monitoring and Maintenance

A system for ongoing monitoring, maintenance, and updates should be established to address any technical issues, bugs, or improvements promptly.

4.2.8 User Training and Onboarding

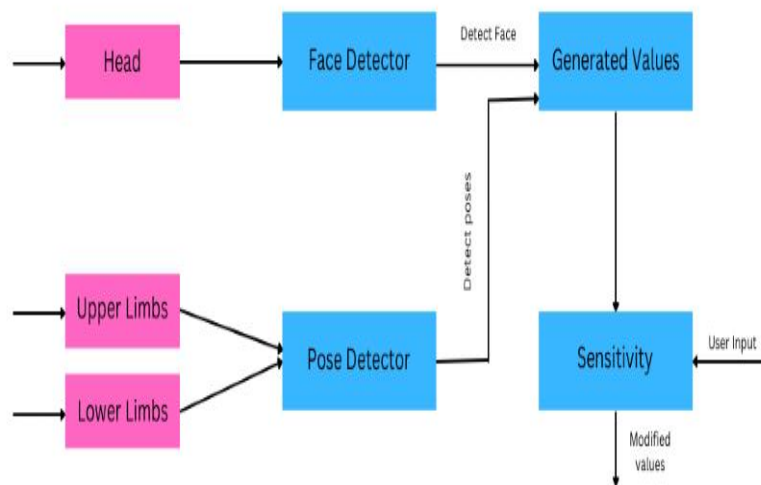
The app should include clear onboarding instructions and tutorials to guide users, caregivers, and healthcare professionals in effectively using the app and maximizing its benefits.

4.2.9 Backup and Data Recovery

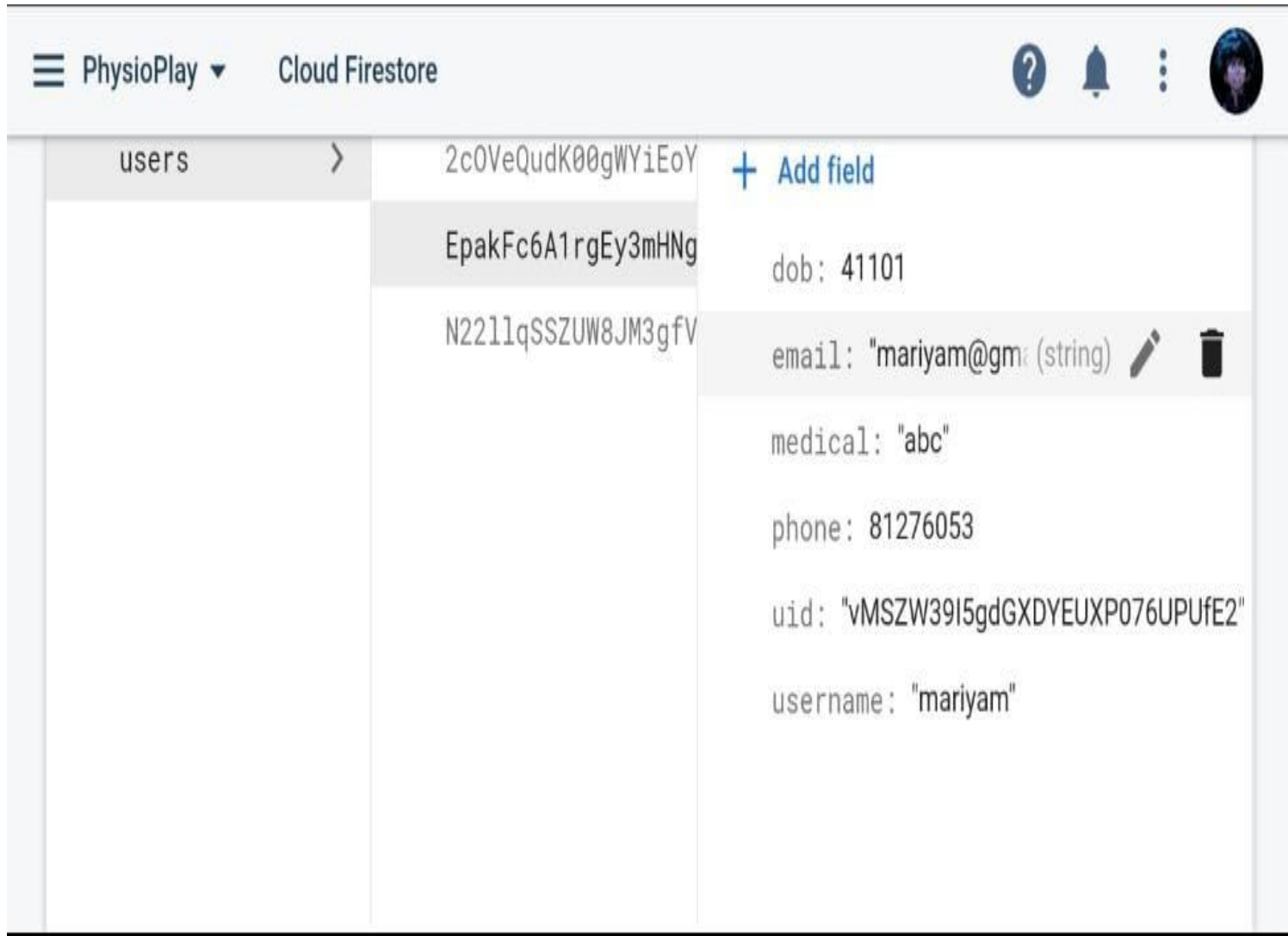
A robust data backup and recovery system should be in place to safeguard user data and allow for swift recovery in case of data loss or system failures.

5. SYSTEM ARCHITECTURE

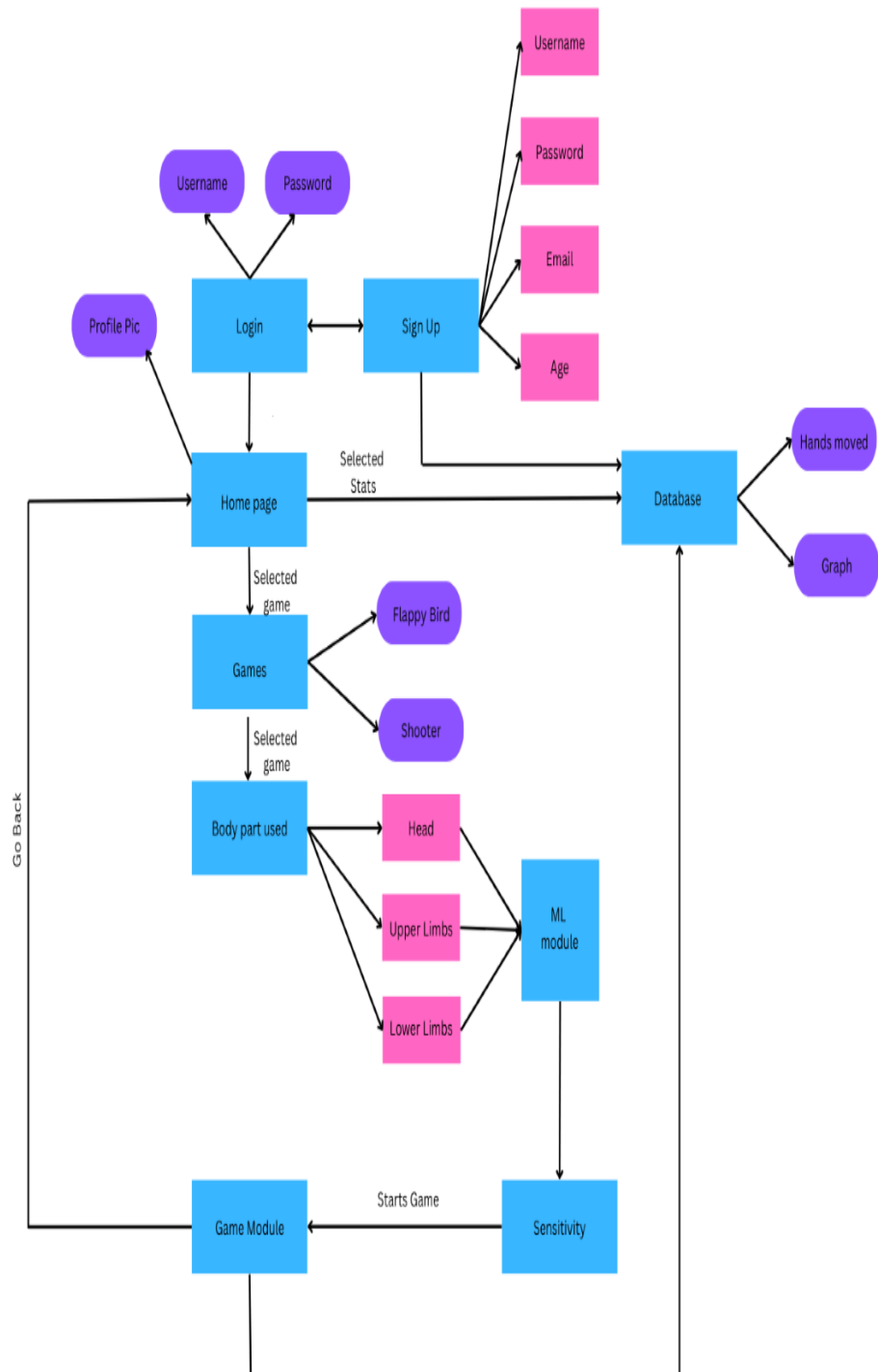
5.1 Machine Learning Architecture



5.2 Database Design



5.3 Flow Control Architecture



6. PROPOSED SYSTEM

6.1 Data Acquisition Module

The Data Acquisition module within the "Physioplay" system serves as a central hub for gathering, processing, and managing crucial user-related data, vital for delivering an effective and personalized physiotherapy experience to children with Autism Spectrum Disorder (ASD) and Cerebral Palsy (CP).

This module encompasses multiple key functionalities that contribute to the app's seamless operation. Firstly, it allows users and caregivers to create personalized profiles, capturing essential information such as age, diagnosis, and therapy objectives. Caregivers can manage profiles for multiple children, ensuring the app tailors exercises to individual needs.

Secondly, the Data Acquisition module facilitates the collection of exercise performance data. During gameplay sessions, it records exercise type, duration, repetitions, and movement accuracy. It integrates with the real-time pose detection component to provide instant feedback on movement correctness, ensuring therapy exercises are executed accurately.

Moreover, the module plays a pivotal role in tracking user progress and achievements. It logs completed exercises, earned rewards, and unlocked levels, fostering motivation and a sense of accomplishment. This progress tracking empowers users and caregivers to monitor developmental milestones and the effectiveness of the therapy program.

The Data Acquisition module also offers a mechanism for user feedback, enabling users and caregivers to provide insights on app usability and potential enhancements. Data privacy and security are paramount, with stringent measures in place to encrypt and protect personal and exercise-related information.

Furthermore, the module serves as a channel for expert collaboration. Physiotherapists and healthcare professionals can review exercise data, offer recommendations, and contribute to personalized therapy plans. This integration ensures that therapy exercises remain aligned with evidence-based practices.

Lastly, the module supports offline data collection, enabling users to engage in exercises even without internet connectivity. Data is synchronized with the central system once a connection is restored.

In summary, the Data Acquisition module plays an indispensable role in facilitating personalized, engaging, and secure physiotherapy experiences within the "Physioplay" system. By capturing and managing user data effectively, it contributes to accurate progress tracking, real-time feedback, user motivation, expert collaboration, and ultimately, the improved well-being of children with ASD and CP.

6.2 Preprocessing Module

The Preprocessing module is a foundational element within the "Physioplay" system, devoted to optimizing the raw exercise performance data collected from children with Autism Spectrum Disorder (ASD) and Cerebral Palsy (CP). Its intricate process involves several key functions that refine and enhance the quality of data before it informs subsequent stages. Through data cleaning and filtering, the module eliminates noise and outliers, ensuring that only accurate and relevant information is retained. Normalization and scaling techniques establish a consistent baseline for data comparison, while time-series alignment synchronizes sequences of movements for precise analysis.

Additionally, the Preprocessing module performs feature extraction to distill meaningful insights from the processed data, capturing essential metrics that illuminate exercise performance. It adeptly handles missing data by employing imputation strategies, ensuring a

comprehensive and complete dataset for evaluation. Quality control mechanisms scrutinize data for anomalies, fostering data reliability and integrity. By seamlessly integrating with the real-time pose detection component, the Preprocessing module contributes to the accuracy of instant feedback provided during therapy exercises, enhancing the overall effectiveness of the app.

In summary, the Preprocessing module's meticulous data refinement process is pivotal to the success of the "Physioplay" system. It transforms raw data into valuable insights, enabling precise analysis, visualization, and personalized feedback. By optimizing the data that informs therapeutic interventions, the Preprocessing module elevates the quality and impact of the physiotherapy experience for children with ASD and CP, exemplifying the system's commitment to innovation and effectiveness.

6.3 Pose Detection Module

Nestled at the heart of the "Physioplay" system, the Pose Detection module stands as a technological cornerstone, leveraging the forefront of machine learning advancements to analyze and interpret the real-time movements of children with Autism Spectrum Disorder (ASD) and Cerebral Palsy (CP) during therapy exercises. This module employs sophisticated pose estimation algorithms that continuously scrutinize users' body postures and movements, accurately tracking joint positions and skeletal configurations. Through this intricate analysis, the module ensures an in-depth comprehension of users' actions.

One of the key strengths of the Pose Detection module lies in its ability to provide instantaneous feedback. Using the real-time movement data, the module generates immediate guidance that assists users in maintaining proper posture and executing exercises with precision. This dynamic feedback mechanism is seamlessly integrated into the app's interface, offering users the

opportunity to make on-the-fly adjustments and cultivate correct movement patterns as they engage in their therapy routines.

Beyond real-time feedback, the Pose Detection module performs a comprehensive assessment of movement quality. It evaluates factors like alignment, symmetry, range of motion, and coordination, contributing to a holistic understanding of users' exercise performance. Furthermore, the module showcases adaptive capabilities, tailoring its guidance to individual needs and developmental levels. In instances where incorrect postures or movements are detected, the module intervenes with corrective strategies, which may involve visual cues, audio prompts, or haptic feedback. This real-time correction empowers users to optimize their exercise execution and refine their movements.

Integral to the overall system, the Pose Detection module seamlessly integrates its processed movement data with other components, ensuring a synchronized and cohesive user experience. As technology evolves and research progresses, the module remains dynamic, adapting and refining its algorithms to sustain a high level of accuracy and reliability. Ultimately, this module not only enhances the effectiveness of therapy exercises but also fosters engagement, empowerment, and meaningful progress for children with ASD and CP within the realm of the "Physioplay" app.

6.4 Game Design Module

Embedded within the intricate tapestry of the "Physioplay" system, the Game Design module stands as a dynamic forge of creativity, meticulously crafting an array of interactive experiences designed to seamlessly intertwine therapeutic goals with captivating gameplay. This module operates at the crossroads of innovative game mechanics and therapeutic principles, resulting in a diverse library of games that cater to the unique needs of children with Autism Spectrum Disorder (ASD) and Cerebral Palsy (CP).

At its core, the module masterfully weaves therapeutic gamification, where each game is meticulously designed to target specific motor skills, coordination, and cognitive development objectives. Gamification elements, including rewards, achievements, and progress tracking, infuse a sense of accomplishment into the gaming journey, motivating users to actively participate in their physiotherapy routines.

The Game Design module leverages interactive gameplay mechanics to bridge the gap between physical movement and in-game outcomes. This design approach encourages users to actively engage in exercises while immersing themselves in the captivating digital world. The module further ensures that sensory considerations are taken into account, creating a harmonious gaming environment tailored to the sensory sensitivities often associated with ASD and CP.

With an unwavering commitment to diversity, the module crafts an assortment of games that encompass varying challenges and activities, ranging from rhythm-based engagements to strategic puzzles. A sophisticated difficulty progression keeps users engaged and invested, ensuring that challenges align with their developmental stages and skill enhancements.

Crucially, the module seamlessly interfaces with the Pose Detection component, harmonizing real-time movement feedback with gameplay. This integration not only cultivates precise exercise execution but also enriches the gaming experience, amplifying the synergy between physical activity and immersive play.

Driven by data insights, the Game Design module continually optimizes its creations. It tailors games to individual preferences through customizable settings, incorporating visual aesthetics and storytelling elements that captivate users' imaginations. By expertly blending therapeutic intention with interactive enjoyment, the module crafts an enriching ecosystem of physiotherapy experiences within the "Physioplay" system, championing engagement, progress, and the well-being of children with ASD and CP.

6.5 Game Control Module

Central to the dynamic ecosystem of the "Physioplay" system, the Game Control module orchestrates a seamless connection between users and the therapeutic games, offering an intuitive and engaging platform for children with Autism Spectrum Disorder (ASD) and Cerebral Palsy (CP) to actively participate in their physiotherapy sessions. This module operates at the forefront of interaction design, enabling users to navigate and influence the game environment through their physical movements and gestures.

At its core, the Game Control module harnesses motion-based control mechanisms, translating users' real-world movements into meaningful in-game actions. By detecting and interpreting distinct gestures, such as reaching, swiping, and tilting, the module transforms the user's physical interactions into a dynamic digital experience, fostering a direct link between exercise execution and gameplay outcomes.

Recognizing the diversity of user capabilities and preferences, the module ensures inclusivity through accessible interfaces. It offers a spectrum of control options, from touch-based interactions to voice commands and switch control. This adaptability caters to users with varying levels of mobility, ensuring that each individual can engage comfortably and effectively.

Incorporating real-time responsiveness as a cornerstone, the module minimizes any delay between users' movements and the corresponding in-game reactions. This immediate feedback loop not only enhances user engagement but also promotes accurate and consistent exercise execution, contributing to the effectiveness of the therapeutic experience.

The Game Control module extends its impact further through multi-modal feedback. By combining visual cues, auditory prompts, and haptic sensations, the module creates a comprehensive feedback ecosystem that guides users toward correct movements and enhances their connection with the game environment.

Embracing customization, the module empowers users and caregivers to calibrate control settings based on individual preferences and capabilities. This personalized approach ensures

that each user can interact comfortably and confidently, tailoring the experience to their specific needs.

Collaborating harmoniously with the Game Design and Pose Detection components, the Game Control module adapts control mappings to align with the mechanics of different games. It seamlessly integrates the physical and digital realms, enabling users to navigate challenges, solve puzzles, and achieve milestones in a manner that resonates with their unique abilities.

In essence, the Game Control module is the conduit that bridges the gap between the physical and virtual worlds, fostering an environment where movement, engagement, and therapeutic progress harmonize seamlessly. By infusing purposeful interaction design with adaptability, responsiveness, and inclusivity, this module empowers children with ASD and CP to embark on a rewarding journey of discovery and well-being within the captivating realm of "Physioplay."

6.6 Evaluation and Analytics Module

Embedded within the fabric of the "Physioplay" system, the Evaluation and Analytics module assumes the role of an astute observer and strategic analyst. Its primary mission revolves around evaluating user progress, assessing the efficacy of therapy interventions, and extracting profound insights from the wealth of data generated during interactive physiotherapy sessions tailored for children with Autism Spectrum Disorder (ASD) and Cerebral Palsy (CP).

Fundamentally, the module meticulously tracks and records users' performance metrics, encompassing exercise completion rates, movement accuracy, and other pertinent parameters. This comprehensive dataset offers a quantitative glimpse into the landscape of therapy engagement and advancement. Through its analytical prowess, the module delves deeper, dissecting the therapeutic outcomes of the exercises, meticulously connecting in-game actions to tangible progress in motor skills, coordination, and cognitive development. By grounding assessments in evidence-based correlations, the module sheds light on the impact of therapy in a manner that resonates beyond mere numbers.

To enhance comprehension and engagement, the module employs adept data visualization techniques. Through these visual representations—ranging from progress charts and graphs to insightful reports—users, caregivers, and healthcare professionals gain an intuitive grasp of developmental milestones and the trajectory of therapy effectiveness. These reports can be uniquely tailored, considering specific exercises, timeframes, and performance markers, thereby enabling personalized analysis and therapeutic planning.

The Evaluation and Analytics module transcends the realm of metrics by extending its purview to user engagement patterns. By scrutinizing trends, preferences, and areas of heightened interest, the module fuels the ongoing evolution of captivating game content, perpetuating sustained user motivation and interaction.

Anchoring the module's capabilities is its potential to inform adaptive therapy plans. Drawing wisdom from the amassed data, it suggests exercise modifications, recalibrates difficulty levels, and recommends personalized game choices, fostering a synergy between therapy and individual requirements.

As a custodian of historical user data, the module allows for longitudinal tracking, facilitating the identification of trends over time. This provides a dynamic perspective for both users and healthcare practitioners to assess progress, make informed decisions, and iteratively adapt therapy strategies for optimal outcomes.

The Evaluation and Analytics module extends its influence beyond immediate application, potentially contributing to scientific exploration and progress. By aggregating anonymized data, it lays the foundation for research initiatives that illuminate the broader impact of interactive therapy on children with ASD and CP, thereby advancing the field of therapeutic interventions. In essence, the module is not solely a repository of data but a generator of actionable insights. By transforming raw information into a compass for progress, the Evaluation and Analytics module empowers stakeholders to navigate the intricacies of therapy with clarity, ultimately nurturing the well-being of children with ASD and CP and contributing to the ongoing refinement of the "Physioplay" system.

7. CODE AND CODE OUTLINE

7.1 Facedetection

This file is dedicated to face detection functionality within the PhsyioPlay application. It plays a crucial role in enabling head control for the games. Using computer vision and facial recognition techniques, this module identifies and tracks the user's face in real-time. The detected facial movements and positions are then translated into input controls for the games. This allows children with cerebral palsy or autism to interact with the games by simply moving their heads, providing an inclusive and accessible gaming experience.

7.2 Upperposedetection

This file focuses on upper pose detection within the PhsyioPlay application. It serves as a key component for controlling games that require upper limb movements. Using computer vision and pose estimation algorithms, this module identifies and tracks the user's upper body and limb positions in real-time. The detected upper limb movements and poses are then translated into input controls for the games. This feature enables children with cerebral palsy or autism to actively engage in games by using their upper limbs, promoting physical exercise and motor skill development while having fun.

7.3 Lowerposedetection

This file is dedicated to lower pose detection within the PhsyioPlay application. Its primary purpose is to provide input controls for games that rely on lower limb movements. Using computer vision and pose estimation techniques, this module identifies and tracks the user's lower body and limb positions in real-time. The detected lower limb movements and poses are then translated into input commands for the games. This functionality enables children with cerebral palsy or autism to actively participate in games by utilizing their lower limbs, fostering physical activity and improving lower body coordination and strength through gameplay.

7.4 processImage()

The `processImage()` function is a critical component of the PhsyioPlay application. It is responsible for calculating coordinates and angles that are essential for the upper pose detection, lower pose detection, and face detection files mentioned earlier. This function takes as input a captured image or frame from the device's camera feed, processes it using computer vision algorithms, and extracts relevant information such as facial landmarks, upper body pose, and lower body pose.

For face detection, `processImage()` identifies key facial features, including the eyes, nose, and mouth, and calculates their positions and movements within the image. This data is then used to determine the user's head movements, which are translated into game controls.

For upper pose detection, the function identifies the positions and orientations of the user's upper limbs, shoulders, and arms. It computes angles and coordinates that represent the upper body's posture, aiding in controlling games that require upper limb movements.

For lower pose detection, `processImage()` tracks the positions and orientations of the user's lower limbs, including the hips, knees, and feet. It calculates angles and coordinates that capture the lower body's posture, enabling control of games that involve lower limb movements.

The `processImage()` function plays a pivotal role in making PhsyioPlay accessible and interactive for children with cerebral palsy or autism, as it allows them to control games through natural body movements, promoting physical exercise and therapeutic benefits.

7.5 welcome.dart

The `welcome.dart` file serves as the initial landing page of the PhsyioPlay application, offering users a straightforward interface with two primary buttons: "Login" and "Signup." This minimalist approach streamlines the user experience by focusing on the essential actions users can take upon launching the app.

Here's a brief overview of the main elements and functionality of this simplified main page

1. **Login Button:** The "Login" button is prominently displayed on the page, typically with clear text labelling. When users tap on this button, they are directed to the login screen, where they can enter their credentials to access their existing accounts.
2. **Signup Button:** The "Signup" button is also prominently displayed, typically adjacent to the login button. When users tap on this button, they are directed to the signup screen, where they can create a new account by providing their information and choosing a username and password.
3. **Minimalistic Design:** The main page typically features a clean and minimalistic design, with a focus on these two essential buttons. This design choice reduces distractions and helps users quickly make a decision to either log in or sign up.
4. **App Logo:** The app's logo and name will be displayed at the top of the page for branding and recognition purposes.
5. **User-Friendly Layout:** The layout and arrangement of the buttons are user-friendly and intuitive, ensuring that users can easily find and select their desired action.

Overall, this minimalistic welcome.dart design prioritizes user simplicity and encourages users to take one of the two primary actions: logging in if they have an existing account or signing up if they are new users. This approach provides a clear and straightforward entry point into the PhsyioPlay application.

7.6 drawerlist.dart

The drawerlist.dart file is a critical component of the PhsyioPlay application, managing the app's navigation and user experience after successful login. It contains a HomeScreen() widget that serves as the main hub of the application for authenticated users. Key elements and functionality include:

1. **User Information:** Displaying the user's name and any relevant profile details.
2. **Progress Tracking:** Showing the user's progress in exercises or games, including scores, achievements, and completion statistics.

3. Game Categories: Organizing games into categories, such as "Head Movements," "Upper Limb Exercises," and "Lower Limb Exercises," providing easy navigation to specific game sections.
4. Log Out Option: Allowing users to securely log out of their accounts when they choose to.

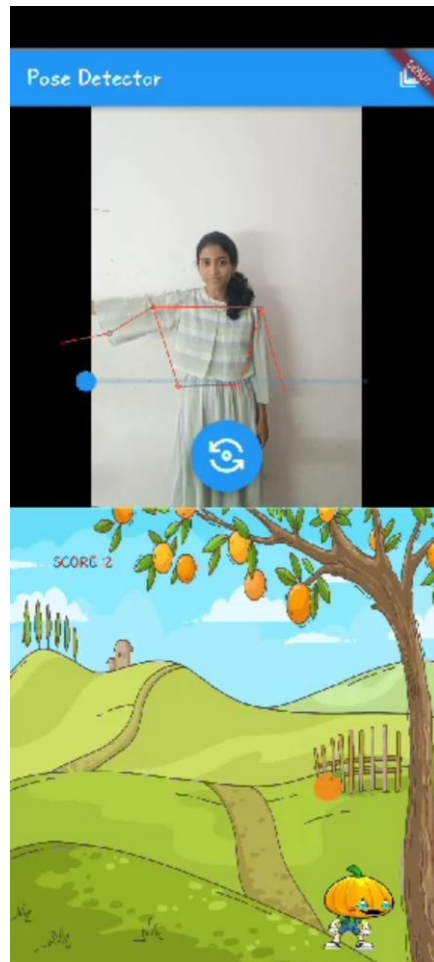
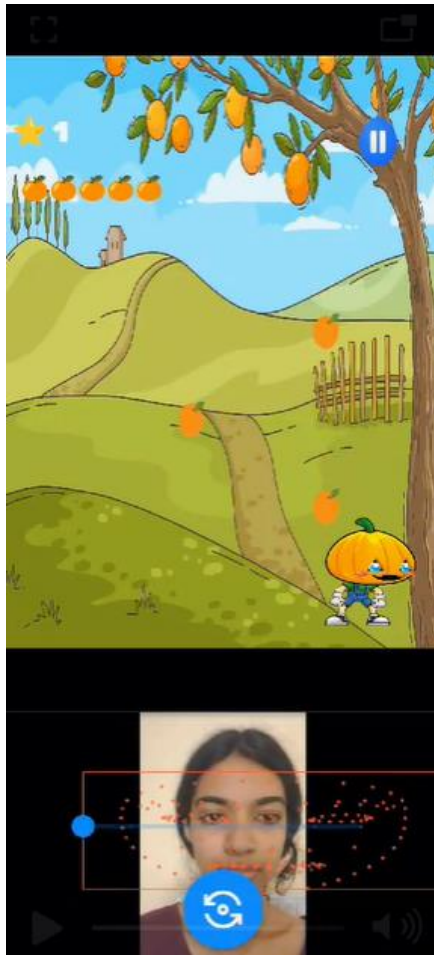
This file ensures a user-friendly and efficient navigation experience for users while offering access to personalized progress data and therapy-specific game categories.

8. OUTPUT SCREENS

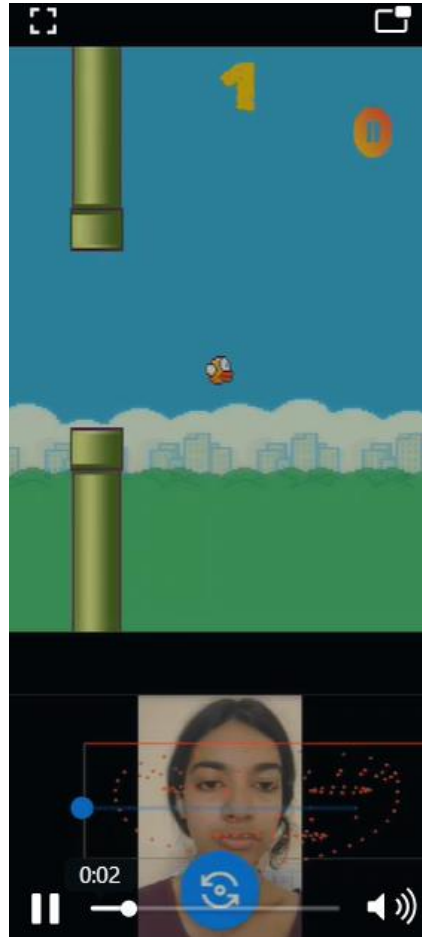
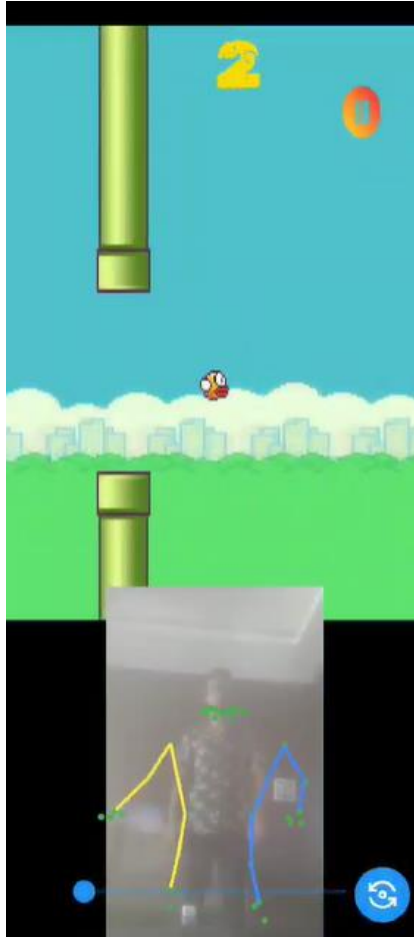
7.1. Home Page and Sign Up page

The image displays two side-by-side mobile app screens. The left screen is the Home Page, featuring the 'PHYSIOPLAY' logo at the top left. Below the logo is an illustration of two people interacting with a large game controller. At the bottom of the screen are two large, rounded buttons: 'LOGIN' (teal) and 'SIGNUP' (cyan). The right screen is the Sign Up page, titled 'SIGNUP' at the top right. It features an illustration of a person standing next to a computer monitor. Below the illustration is a form with the following fields: 'Your Email :', 'Username :', 'Password :', 'Confirm Password :', 'Age :', and 'Medical Condition :'. Each field has a corresponding icon (person, lock, or eye). Below the form is a large teal 'SIGNUP' button. At the bottom of the sign-up page, there is a link that says 'Already have an Account ? Sign In'. A home icon is visible in the bottom right corner of the sign-up page. The top of both screens shows a status bar with the time 1:07, signal strength, and battery level at 71%.

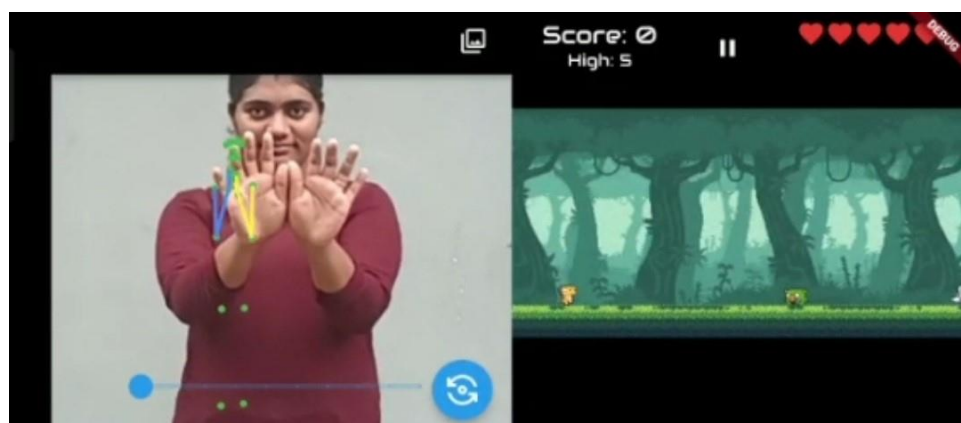
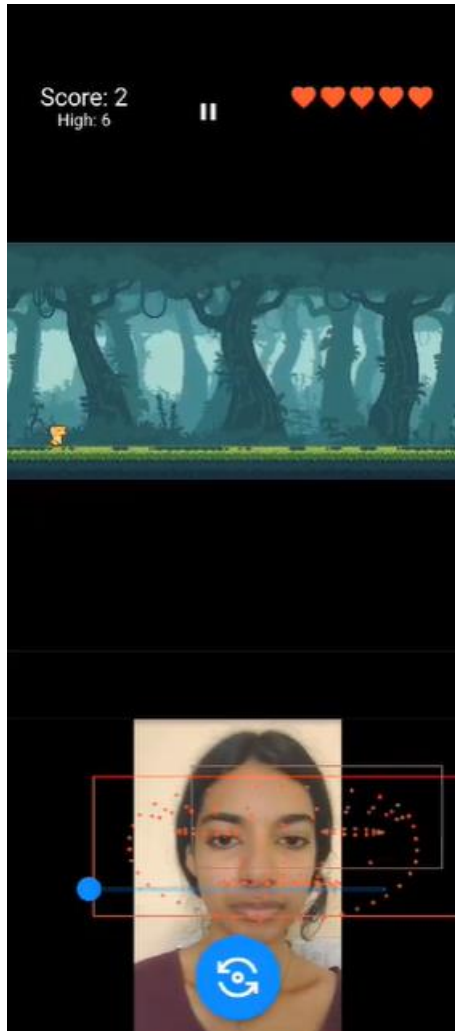
8.2. Fruit Collection



8.3. Flappy Bird



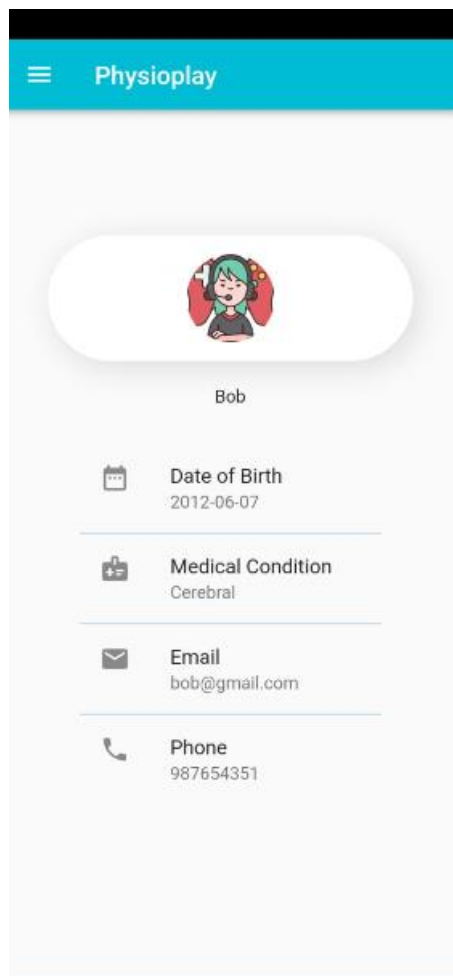
8.4. Dino Run



8.5. Quiz



8.6. Dashboard



9. CONCLUSION AND FURTHER WORK

The culmination of the "Physioplay" system signifies a pioneering leap at the nexus of technology and therapy, embracing the noble mission of elevating the lives of children with Autism Spectrum Disorder (ASD) and Cerebral Palsy (CP). The intricate interplay of its innovative components forms a comprehensive platform that fuses engagement with effective physiotherapy interventions.

Through seamless orchestration, each module contributes to a symphony of innovation. From capturing essential data in the Data Acquisition module to refining it in the Preprocessing module, and further integrating real-time movement analysis through the Pose Detection module, the foundation is set. This harmony continues with the Game Design module crafting motivating experiences and the Game Control module enabling intuitive interactions. Guiding the cadence, the Evaluation and Analytics module transforms data into insightful progress, fostering evidence-based decisions.

The "Physioplay" system is a testament to the potential of technology in fostering positive change. By nurturing engagement, fostering progress, and infusing a sense of accomplishment, it reshapes therapy's boundaries. As this symphony of innovation concludes, the resonance of progress stands as an enduring testament, illuminating a path of hope and inspiration. The

"Physioplay" system harmoniously harmonizes to compose a transformative melody, propelling therapeutic empowerment for children with ASD and CP into an enlightened future.

Further Work

- i. **Enhanced Game Library:** Continuously enriching the collection of therapeutic games by incorporating diverse challenges and activities can sustain user engagement and cater to evolving therapeutic needs. Collaborations with game designers and therapists can bring fresh insights and creative gameplay mechanics.
- ii. **Multilingual Support:** Offering multilingual interfaces and exercises can make the system more accessible to a global audience, ensuring that children from diverse linguistic backgrounds can benefit from its therapeutic offerings.
- iii. **Wearable Integration:** Integrating with wearable devices could enhance movement tracking accuracy and enable users to engage in exercises beyond the app, promoting a holistic approach to therapy.
- iv. **Continuous User Feedback:** Implementing mechanisms for users and caregivers to provide feedback and suggestions can facilitate ongoing improvements, ensuring that the system remains responsive to user needs.

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11 APPENDIX-I

1.

INTRODUCTION

Physioplay is an Android game developed for use in physical rehabilitation and therapy programs to help patients improve their physical function, mobility, and overall health. The game will use various game mechanics and interactive features to engage and motivate patients to perform exercises and movements prescribed by their physiotherapists. The PhysioPlay App is designed for users who require head ,upper and lower limb physiotherapy, providing them with games and feedback for the therapy , helping them achieve their health goals.

DOCUMENT PURPOSE

The aim of this document is to specify the software requirements for a gaming app that utilises pose detection of the face and limbs for gameplay. The app will be built using the Flutter framework and will incorporate features such as user registration, various games utilising pose detection, and score tracking. The purpose of the app isto provide users with an engaging and interactive gaming experience that promotes physical movement and activity.

PRODUCT SCOPE

This app is designed to serve two main purposes. Firstly, it is intended for individuals who require body movement physiotherapy as part of their rehabilitation and recovery process. Secondly, it is suitable for individuals who wish to maintain their overall health and fitness through engaging in physical activity. The app will be user-friendly, responsive, and secure, offering a seamless user experience. It will be optimised to run smoothly on Android devices, providing a high level of accessibility to users.

INTENDED AUDIENCE AND DOCUMENT OVERVIEW

Readers include the panel of professors and the team members consisting of developers and documentation writers. The document consists of a detailed description of all the functional and non-functional requirements and various perspectives of the product. It has all the features that are to be implemented in the product.

STAKEHOLDERS

- PATIENTS
- DOCTORS
- PARENTS OR CAREGIVERS
- EDUCATORS AND THERAPISTS
- ADVOCACY ORGANISATIONS

DOCUMENT CONVENTIONS

Use Times New Roman font for the entire report – Chapter/Section Title – Times New Roman 18, Bold; Heading 2 – Times New Roman 16, Bold; Heading 3 – Times New Roman 14, Bold; Body- Times New Roman 13, Normal.

- Line Spacing – Between Heading 2 – 3 lines, between lines in paragraph 1.5 lines.
- Alignments – Chapter/Section Title – Center, Heading 2 & 3 should be Left Aligned. Ensure that all body text is paragraph justified.
- Figures & Tables – Ensure that all Figures and Tables are suitably numbered and given proper names/headings. Write figure title under the figure and table title above the table.

REFERENCES AND ACKNOWLEDGMENTS

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2.

OVERALL DESCRIPTION

PRODUCT OVERVIEW:

Product Perspective: The mobile application will provide a novel approach to physiotherapy for individuals with autism and cerebral palsy by using games controlled by body movements to promote mobility and range of motion. The application will be developed using ML Kit in Flutter, enabling the integration of machine learning algorithms to detect and respond to user movements in real-time.

PRODUCT FEATURES:

- Games controlled by body movements that promote mobility and range of motion
- Customizable and adaptive games that cater to the user's needs and abilities
- Fun and interactive game experiences that engage the user in physiotherapy
- Machine learning algorithms that detect and respond to user movements in real-time
- Data collection features to track progress and user engagement

USER CLASSES AND CHARACTERISTICS

The target user group for the application is individuals with autism and cerebral palsy who require physiotherapy. The users are expected to have varying levels of mobility and cognitive function. Another user group is therapists who can add patient information, choose appropriate options and track their progress. Operating Environment: The mobile application will be developed for Android devices. The application will require an internet connection for data collection and storage.

DESIGN AND IMPLEMENTATION:

- The application is designed to work on both iOS and Android devices.
- The Android API level should be above 21.
- The design and implementation of the device should prioritise the inclusion of a high-quality camera, capable of meeting the requirements of its intended purpose and target audience.
- The application must comply with the General Data Protection Regulation (GDPR)
- The application must be developed using Flutter

- The user interface should be simple and easy to understand, with readable text and smooth transitions.
- The application should be space-efficient and use a readable and maintainable code.

- The variables should be named using camel case, and the class names should start with a capital letter.
- The application should follow proper exception handling mechanisms
- The front-end should be developed using Dart programming language

ASSUMPTIONS AND DEPENDENCIES :

The assumptions are:-

- The system meets all the hardware requirements
- The user has internet browsing capability and an internet connection
- The system should have more capacity and provide fast access to the database
- The games will be developed for both Android and iOS platforms, with similar gameplay and user interface across both platforms
- The games will be designed to be accessible and enjoyable for people with different levels of cognitive and physical abilities, without causing frustration or sensory overload

The dependencies are:-

- The specific hardware and software due to which the product will run
- The end users should have proper understanding of the product
- On the basis of listing requirements and specifications the project will develop and run
- The information of all users must be stored in a database that is accessible by the system
- The face-tracking technology may require customization or calibration to work effectively with the target users, which may require additional testing and development time

3.

SPECIFIC REQUIREMENTS

EXTERNAL INTERFACE REQUIREMENTS:

User Interfaces

- A. Sign in page
- B. Login page
- C. Profile page
- D. Games page
- E. Difficulty choosing page

Hardware Interfaces

- A. Operating system: android
- B. Disk Storage: 4 GB RAM : 128 GB SSD
- C. Processor : Qualcomm Snapdragon 845 and above

Software Interfaces

- A. Firebase
- B. Figma
- C. App Store
- D. GitHub

FUNCTIONAL REQUIREMENTS:

Use Case	Description
Register	The user can register by typing their credentials
Login	The user can login into their account using correct credentials
Play Game	Any of the games that are offered can be chosen by the user and played using either the selected head or limbs.
Face Point Detection	Real Time detection of user's face point with accuracy to control the game
Track Score	The user score will be captured and displayed then it will be stored in the database. The user can view their scores for each game and compare them over time to monitor their progress.
Adjust Sensitivity	The sensitivity level for each game can be adjusted based on the user's preference. The sensitivity levels will be Low, Medium, and High, and the higher the sensitivity, the faster the avatar will move.
Set Timer	The Fruit Collection game will have a timer of 60 seconds, after which the game will end. The user will be notified when the time is about to run out, and the final score will be recorded.
Notification	The user can receive notifications from the application, reminding them to play games or perform other exercises.

USE CASE MODEL:

1) *Use Case Name: Login*

Summary: The login feature allows the actor of the software to use the application as a registered user.

Preconditions: Create an account.

Triggers: Actor selects the sign in button.

Basic course of events (Scenario):

Internal Precondition: The sign in page is displayed.

Actor	System	Screen
1. The actor selects to sign in		
	2. Email and password	
3. Actor completes fields and select sign in		
	4. The Application validates actor as registered user	

Internal Post Condition:

1. Actor confirmed by the system as a registered user.
2. Customers can proceed to use the website.

ALTERNATIVE PATHS :

Actors sign in as guest users.

Internal precondition: None

Actor	System	Screen
1. Actors sign in as guest users		

Internal Post Conditions:

1. Customers can proceed to select the difficulty level.

2) Use Case Name: Difficulty Level

Summary: The feature allows the actor of the software to choose the difficulty level of the game.

Preconditions: Sign in page is displayed **Triggers:**

Actor selects the difficulty level button. **Basic course of events (Scenario):**

Internal Precondition: The actor signed in using registered username and password.

Actor	System	Screen
	1. Displays the difficulty level page	
2. Actor chooses from easy, medium, hard levels		

	3. The Application validates the level of the game	
--	--	--

Internal Post Conditions:

1. Customers can proceed to select the game.

3) Use Case Name: Selecting Game

Summary: The feature allows the actor of the software to choose the required game to play.

Preconditions: Difficulty level interface is displayed.

Triggers: Actor selects game button.

Basic course of events (Scenario):

Internal Precondition: None.

Actor	System	Screen
	1. Displays the selection page	
2.Actor chooses the required game to play		
	3.The application validates the game	

Internal Post Condition:

1. Users can start the game now.

NON-FUNCTIONAL REQUIREMENTS:

- **Performance:** The app is fast, responsive, and reliable, with minimal loading times and smooth transitions between screens.
- **Usability:** The app is user-friendly, with an intuitive interface that is easy to navigate, understand, and use. It is accessible to users with disabilities.
- **Security:** Security is ensured with strong data encryption and authentication protocols to protect user data and prevent unauthorised access.
- **Compatibility:** Compatible with different mobile platforms, devices, and screen sizes, and works seamlessly with different versions of operating systems.
- **Reliability:** The application is reliable and free from errors or crashes.
- **Maintainability:** It is made easy to maintain and update the app, with clear documentation, modular design, and efficient code.

SYSTEM DESIGN ARCHITECTURE :

The application will be developed using a client-server architecture. The client will be the mobile application, while the server will store user data and game content. Data Flow Diagrams: The data flow diagram will show the flow of data within the system, from the user's interaction with the games to the storage of data on the server.

SOFTWARE QUALITY ATTRIBUTES :

- **Capacity:** The storage requirements of the app can be minimal since no information about the user is stored.
- **Compatibility:** Compatible with any version of the android which supports voice assistants.
- **Response Time:** Each page loads within 10-15 seconds and the output is expected to have the same response time.
- **Reliability:** Chances of failure or system crash is low compared to other heavy applications that require huge storage.
- **Availability:** The app will be always available to the user except times of

maintenance. It will be made available in the Play Store for installation.

- Efficiency: The app is very efficient as the user can reach their goal within minimal time.

